

1. What is discourse in Natural Language Processing?

Answer: Discourse refers to a sequence of related sentences that together communicate a meaningful idea. In NLP, discourse analysis studies how sentences are connected and how meaning is maintained across an entire text or conversation.

2. What is discourse analysis?

Answer: Discourse analysis is the process of studying language beyond individual words and sentences. It examines relationships between sentences, speaker intentions, context, coherence, and the overall structure of communication.

3. What is reference resolution?

Answer: Reference resolution is the process of identifying the entity or object referred to by expressions such as pronouns and noun phrases. For example, in "Ravi bought a book. He read it," the system identifies Ravi as "He" and the book as "it."

4. What is anaphora resolution?

Answer: Anaphora resolution is a type of reference resolution in which a word refers back to a previously mentioned entity. Pronouns such as "he," "she," "it," and "they" commonly require anaphora resolution.

5. What is text coherence?

Answer: Text coherence refers to the logical and meaningful connection between different parts of a text. A coherent text allows readers to understand how ideas and sentences relate to each other.

6. Analyze the reference resolution in: "Anita bought a laptop. She uses it every day."

Answer: The pronoun "She" refers to Anita because Anita is the previously mentioned female entity. The pronoun "it" refers to the laptop, and identifying these references helps the system correctly understand the complete discourse.

7. Analyze the coherence of: "It was raining heavily. Therefore, the match was cancelled."

Answer: The two sentences have a clear cause-and-effect relationship. Heavy rain is the cause, while cancellation of the match is the result, creating a coherent connection between the statements.

8. Analyze the ambiguity in: "John told David that he was selected."

Answer: The pronoun "he" is ambiguous because it could refer to either John or David. Reference resolution requires contextual information to determine which person was actually selected.

9. Analyze the reference in: "The dog chased the cat because it was afraid."

Answer: The pronoun "it" is ambiguous because it may refer to either the dog or the cat. Semantic and contextual reasoning are required to determine which animal was afraid.

10. Analyze the dialog acts in: "Hello. How can I help you? I need information about admissions."

Answer: "Hello" is a greeting, while "How can I help you?" is an offer of assistance. The final statement is an information request, showing multiple dialog acts within a short conversation.